



DSA0602-DATA HANDLING AND DATA VISUALIZATION

SLOT A

Dr. ALMAS BEGUM

PRESENTED BY:

SURU SAI KEERTHI [192424368]

SET 1:

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

Install package (Run only once)

```
install.packages("ggplot2")
```

Load Library

```
library(ggplot2)
```

```
sales <- data.frame(
```

```
  Month = factor(c("January", "February", "March", "April", "May"),
```

```
               levels=c("January", "February", "March", "April", "May")),
```

```
  Sales = c(15000, 18000, 22000, 20000, 23000)
```

```
)
```

```
ggplot(sales, aes(x=Month, y=Sales, group=1))+
```

```
  geom_line(color="blue", linewidth=1.2)+
```

```
  geom_point(color="red", size=3)+
```

```
  geom_text(aes(label=Sales), vjust=-1, size=4)+
```

```
  labs(
```

```
    title="Monthly Sales",
```

```
    x="Month",
```

```
    y="Sales ($)"
```

```
  )+
```

```
  theme_minimal()
```

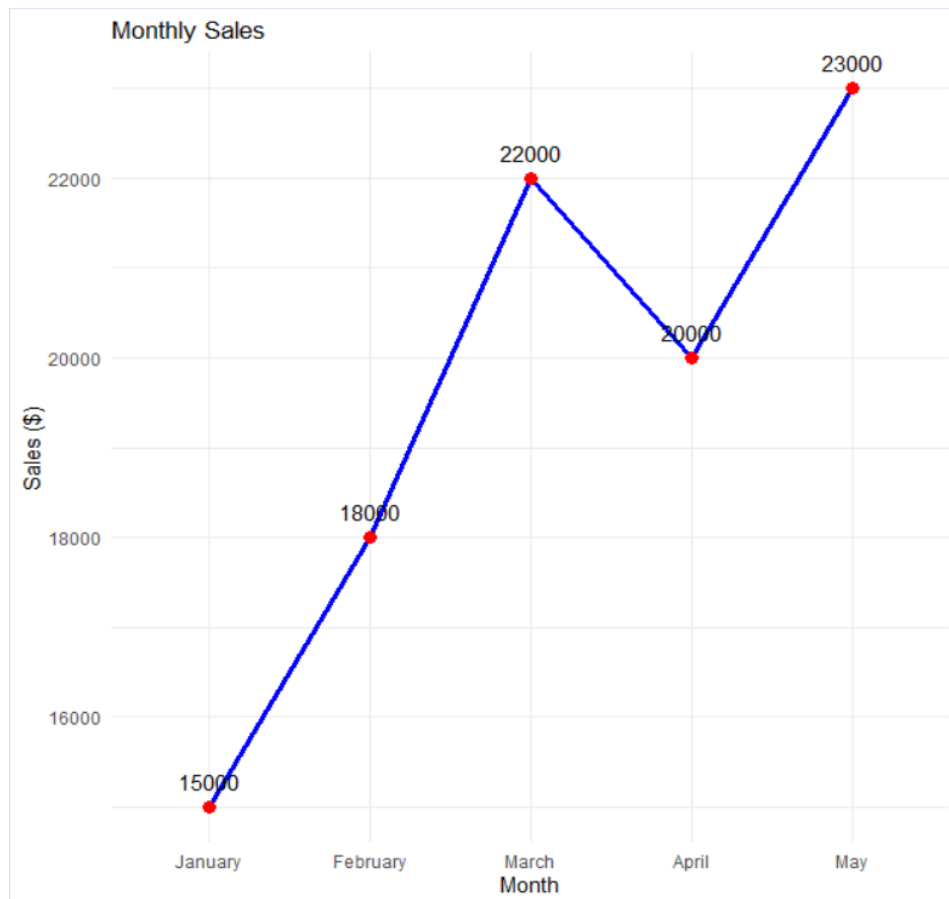
```
#####
```

```
# Dataset 2 : Top Selling Products
```

```
#####
```

```
products <- data.frame(  
  Product=c("Laptop","Mobile","Tablet","Headphones","Smart Watch"),  
  Sales=c(52000,48000,35000,28000,25000)  
)
```

```
ggplot(products, aes(x=Product, y=Sales, fill=Product))+
```



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```

geom_bar(stat="identity")+
  geom_text(aes(label=Sales), vjust=-0.5, size=4)+
  labs(
    title="Top Selling Products",
    x="Products",
    y="Sales ($)"
  )+
  theme_minimal()

#####

# Dataset 3 : Advertising Budget

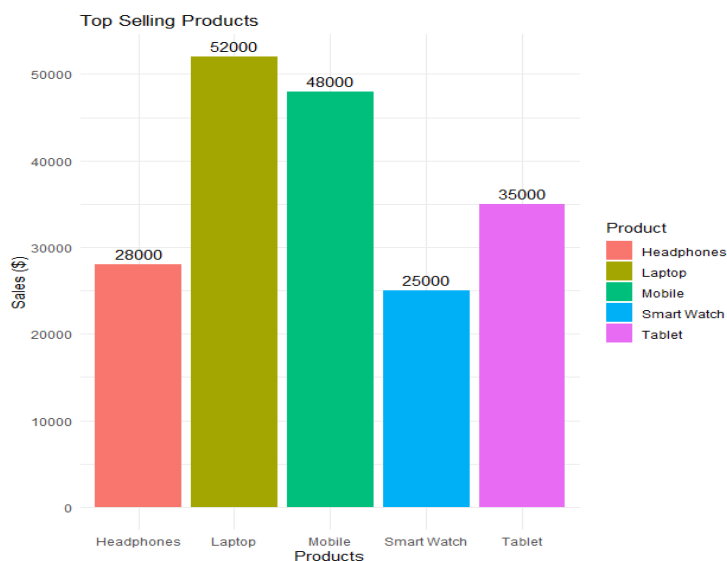
#####

```

```

advertising <- data.frame(
  Budget=c(3000,4000,5000,4500,5500),
  Sales=c(15000,18000,22000,20000,23000),
  Month=c("January","February","March","April","May")
)

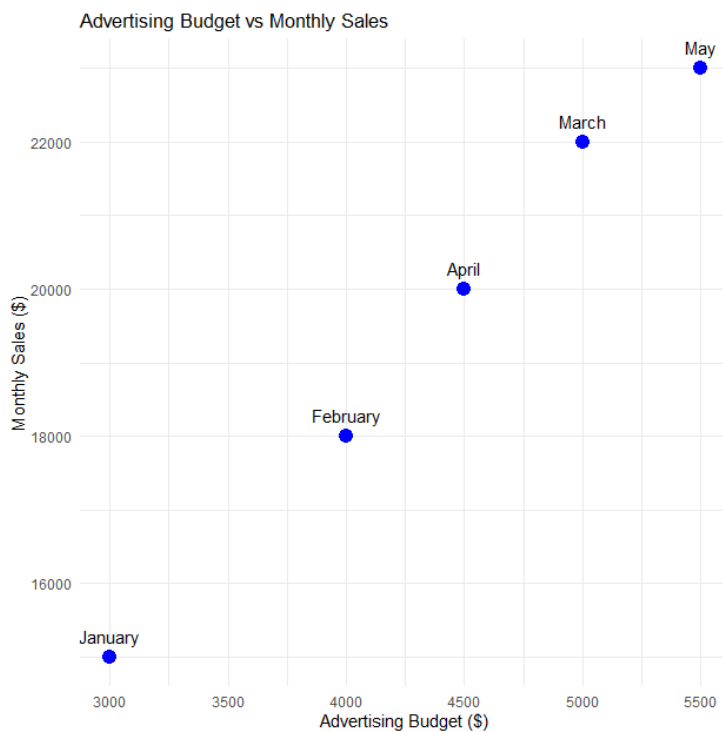
```



3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales

Explain the insights drawn from the scatter plot.

```
ggplot(advertising,  
  aes(x=Budget,  
    y=Sales))+  
geom_point(color="blue", size=4)+  
geom_text(aes(label=Month),  
  vjust=-1,  
  size=4)+  
labs(  
  title="Advertising Budget vs Monthly Sales",  
  x="Advertising Budget ($)",  
  y="Monthly Sales ($)"  
) +  
theme_minimal()  
print("Insight:")  
print("Higher advertising budgets generally lead to higher monthly sales.")
```



4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

```
par(mfrow=c(1,2))
```

```
# Line Chart
```

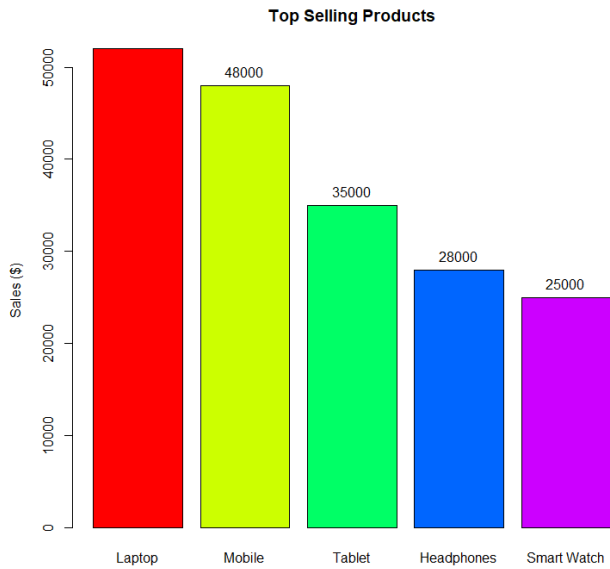
```
plot(  
  sales$Month,  
  sales$Sales,  
  type="o",  
  col="blue",  
  pch=16,  
  xlab="Month",  
  ylab="Sales ($)",  
  main="Monthly Sales"  
)
```

```
text(  
  sales$Month,  
  sales$Sales,  
  labels=sales$Sales,  
  pos=3  
)
```

```
# Bar Chart
```

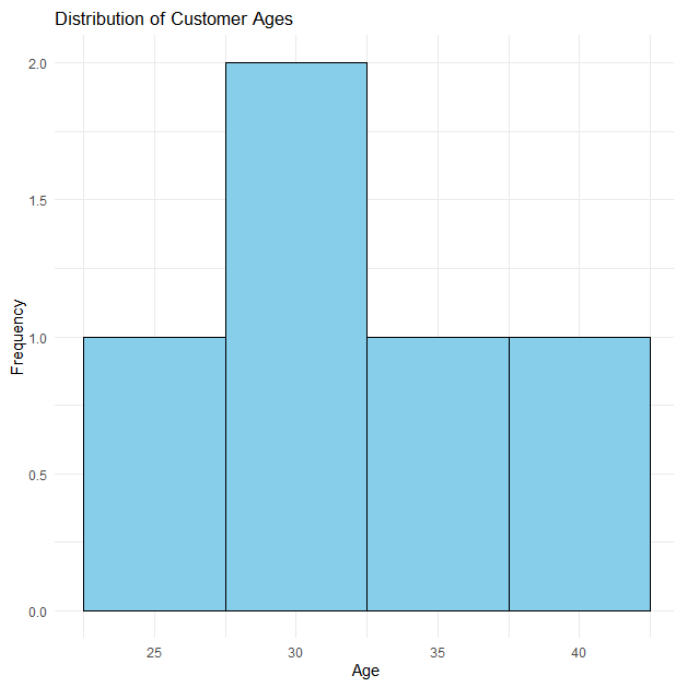
```
bar <- barplot(  
  products$Sales,  
  names.arg=products$Product,  
  col=rainbow(5),  
  main="Top Selling Products",  
  ylab="Sales ($)"  
)
```

```
text(  
  bar,  
  products$Sales,  
  labels=products$Sales,  
  pos=3  
)  
par(mfrow=c(1,1))
```



1.Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
install.packages("ggplot2")
install.packages("wordcloud")
install.packages("RColorBrewer")
library(ggplot2)
library(wordcloud)
library(RColorBrewer)
customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)
ggplot(customer, aes(x=Age)) +
  geom_histogram(
    binwidth=5,
    fill="skyblue",
    color="black"
  ) +
  labs(
    title="Distribution of Customer Ages",
    x="Age",
    y="Frequency"
  ) +
  theme_minimal()
```



2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
satisfaction <- as.data.frame(table(customer$Satisfaction))
```

```
colnames(satisfaction) <- c("Score", "Count")
```

```
ggplot(satisfaction,
```

```
  aes(x="", y=Count, fill=Score)) +
```

```
  geom_bar(stat="identity", width=1) +
```

```
  coord_polar("y") +
```

```
  geom_text(aes(label=Count),
```

```
    position=position_stack(vjust=0.5)) +
```

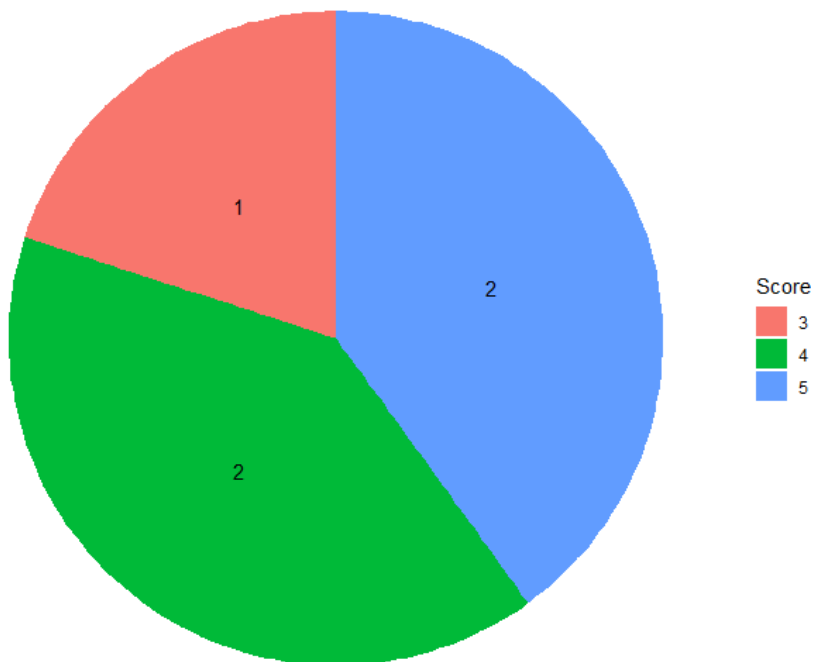
```
  labs(
```

```
    title="Customer Satisfaction Distribution"
```

```
  ) +
```

```
  theme_void()
```


Customer Satisfaction Distribution



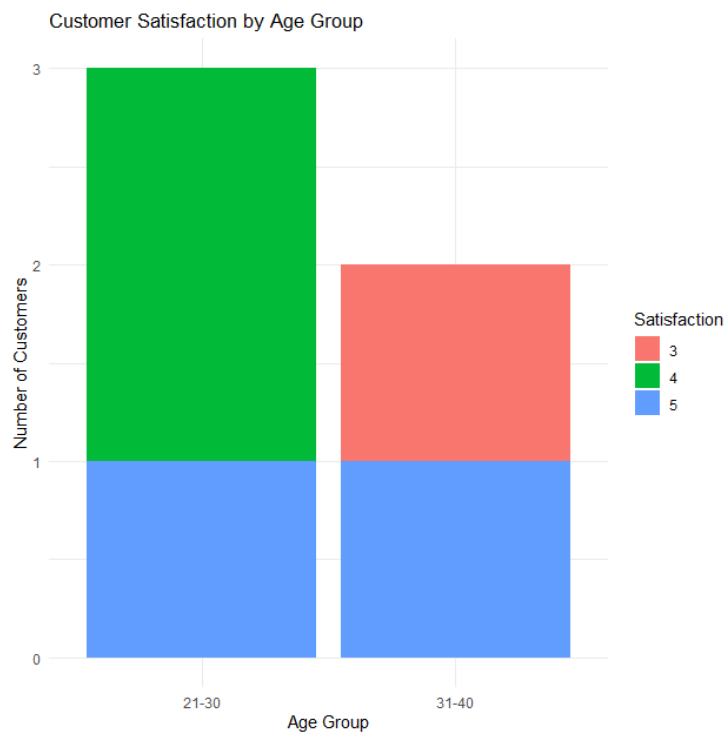
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
customer$AgeGroup <- cut(
  customer$Age,
  breaks=c(20,30,40,50),
  labels=c("21-30","31-40","41-50"),
  include.lowest=TRUE
)

ggplot(customer,
  aes(x=AgeGroup,
      fill=factor(Satisfaction))) +
  geom_bar() +
  labs(
    title="Customer Satisfaction by Age Group",
    x="Age Group",
    y="Number of Customers",
    theme="ggplot2")
```

```
) +
```

```
theme_minimal()
```



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
feedback <- c(
```

```
"Excellent service",
```

```
"Very satisfied",
```

```
"Good quality",
```

```
"Fast delivery",
```

```
"Excellent support",
```

```
"Satisfied customer",
```

```
"Helpful staff",
```

```
"Good experience",
```

```
"Quick response",
```

```
"Amazing service"
```

```
)
```

```
wordcloud(
```

```
words=feedback,
```

```

random.order=FALSE,
colors=brewer.pal(8,"Dark2"),
scale=c(4,1)
)

```



SET3:

```

install.packages("ggplot2")
library(ggplot2)
employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsOfService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)

```

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

```

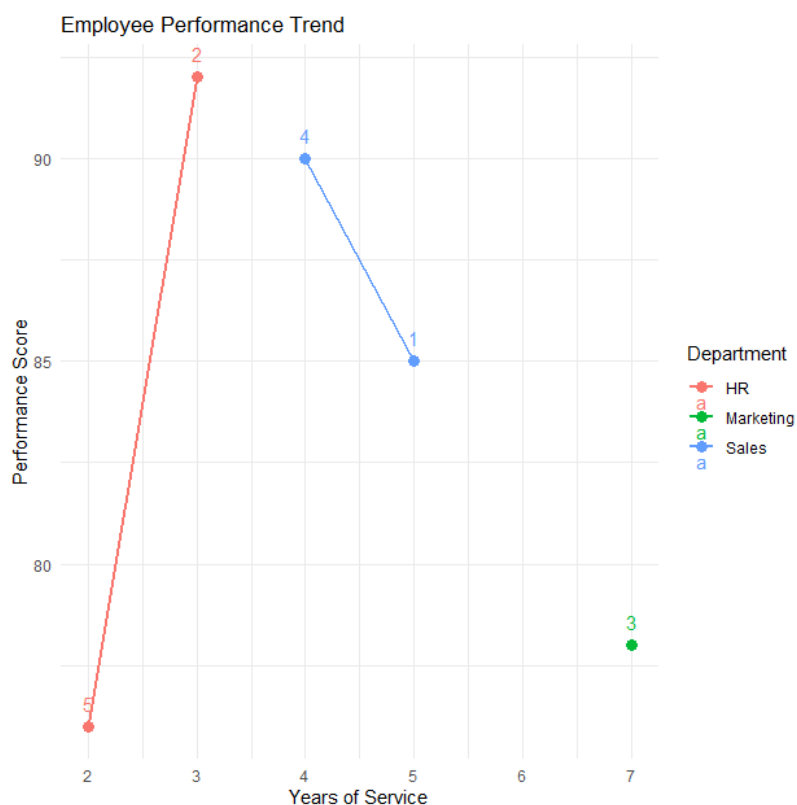
ggplot(employee,
  aes(x=YearsOfService,
    y=PerformanceScore,

```

```

    color=Department,
    group=Department)))+
  geom_line(linewidth=1)+
  geom_point(size=3)+
  geom_text(aes(label=EmployeeID),
    vjust=-1)+
  labs(
    title="Employee Performance Trend",
    x="Years of Service",
    y="Performance Score",
    color="Department"
  )+
  theme_minimal()

```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```

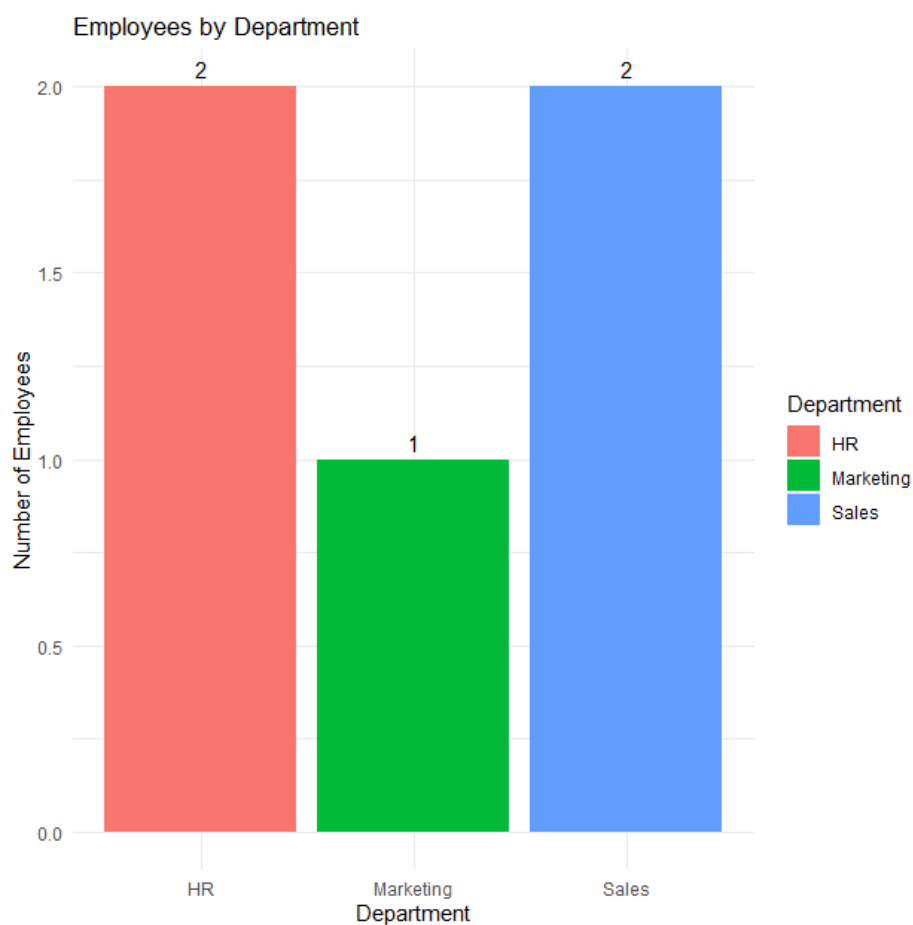
ggplot(employee,
  aes(x=Department,
    fill=Department))+

```

```

geom_bar()+
geom_text(
  stat="count",
  aes(label=after_stat(count)),
  vjust=-0.5
)+
labs(
  title="Employees by Department",
  x="Department",
  y="Number of Employees"
)+
theme_minimal()

```



3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
ggplot(employee,
```

```

aes(x=YearsOfService,
    y=PerformanceScore,
    color=Department))+
geom_point(size=4)+
geom_text(aes(label=EmployeeID),
    vjust=-1)+
labs(
    title="Years of Service vs Performance Score",
    x="Years of Service",
    y="Performance Score"
)+
theme_minimal()
print("Insight:")
print("The scatter plot shows no strong linear relationship between years of service and performance score.")
print("Some employees with fewer years of service have higher performance scores, indicating that experience alone does not determine performance.")

```



4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

```
par(mfrow=c(1,2))
```

```
# Line Chart
```

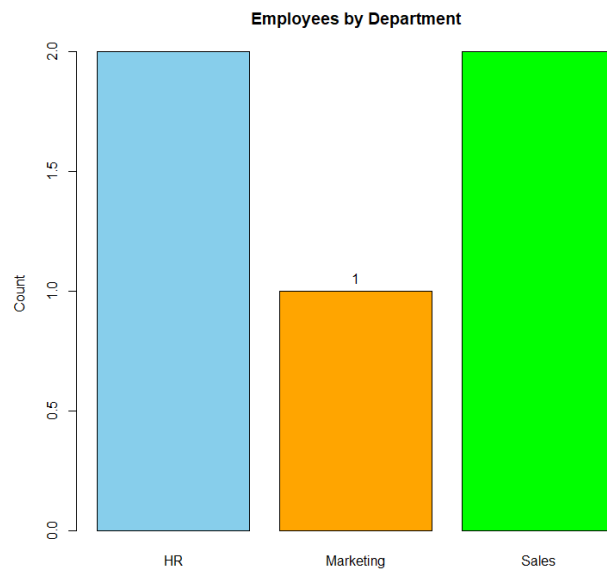
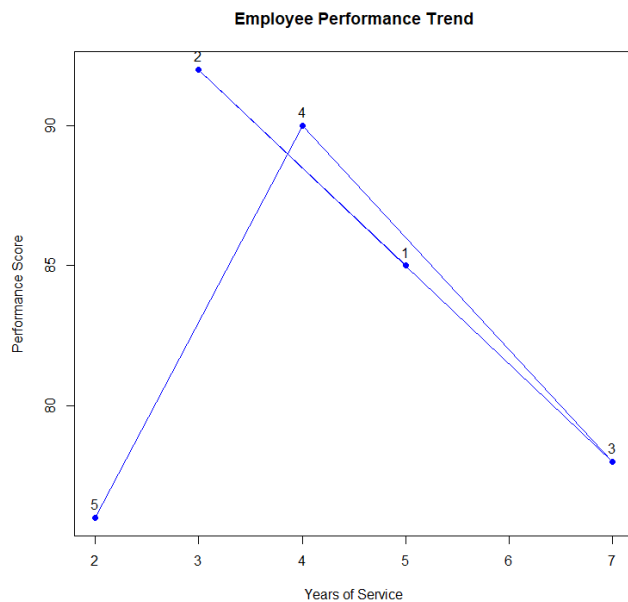
```
plot(  
  employee$YearsOfService,  
  employee$PerformanceScore,  
  type="o",  
  col="blue",  
  pch=16,  
  xlab="Years of Service",  
  ylab="Performance Score",  
  main="Employee Performance Trend"  
)
```

```
text(  
  employee$YearsOfService,  
  employee$PerformanceScore,  
  labels=employee$EmployeeID,  
  pos=3  
)
```

```
bar <- barplot(  
  table(employee$Department),  
  col=c("skyblue","orange","green"),  
  main="Employees by Department",  
  ylab="Count"  
)
```

```
text(  
  bar,  
  table(employee$Department),  
  labels=table(employee$Department),  
  pos=3  
)
```

```
par(mfrow=c(1,1))
```



SET 4:

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
inventory <- data.frame(
```

```
  ProductID = c(1,2,3,4,5),
```

```
  ProductName = c("Product A","Product B","Product C","Product D","Product E"),
```

```
  Quantity = c(250,175,300,200,220),
```

```
  Category = c("Electronics","Electronics","Furniture","Furniture","Accessories"),
```

```
  Price = c(1200,850,1500,950,700)
```

```
)
```

1.Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
ggplot(inventory,
```

```
  aes(x=ProductName,
```

```
    y=Quantity,
```

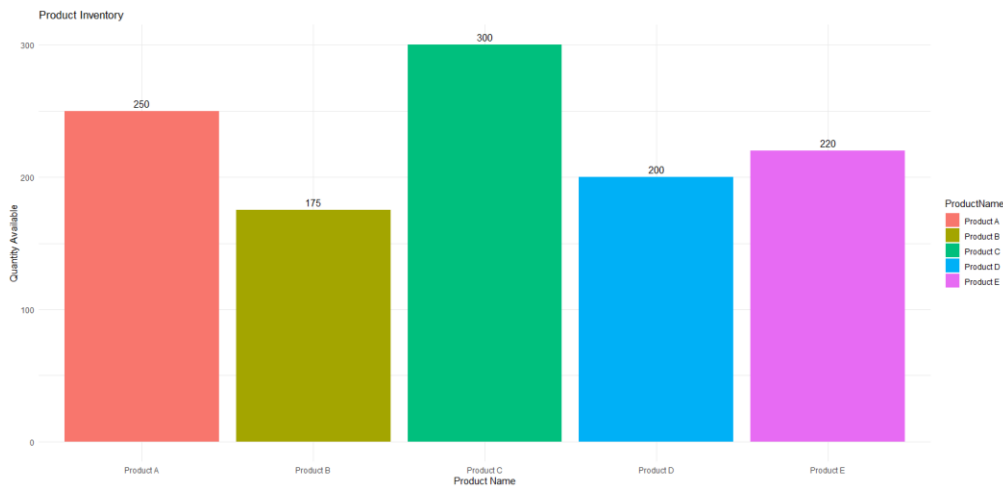
```
    fill=ProductName))+
```

```
geom_bar(stat="identity")+
```



```
geom_text(aes(label=Quantity),
  vjust=-0.5)+
```

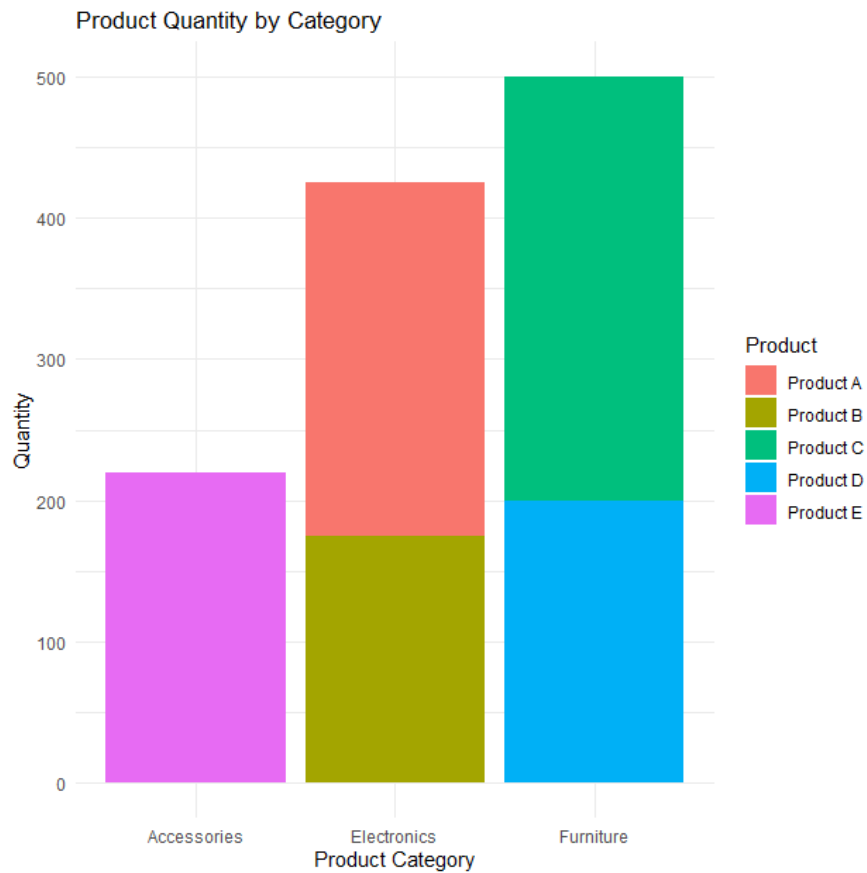
```
labs(
  title="Product Inventory",
  x="Product Name",
  y="Quantity Available"
)+
theme_minimal()
```



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
ggplot(inventory,
  aes(x=Category,
    y=Quantity,
    fill=ProductName))+
geom_bar(stat="identity")+
labs(
  title="Product Quantity by Category",
  x="Product Category",
  y="Quantity",
  fill="Product"
)+
```

```
theme_minimal()
```



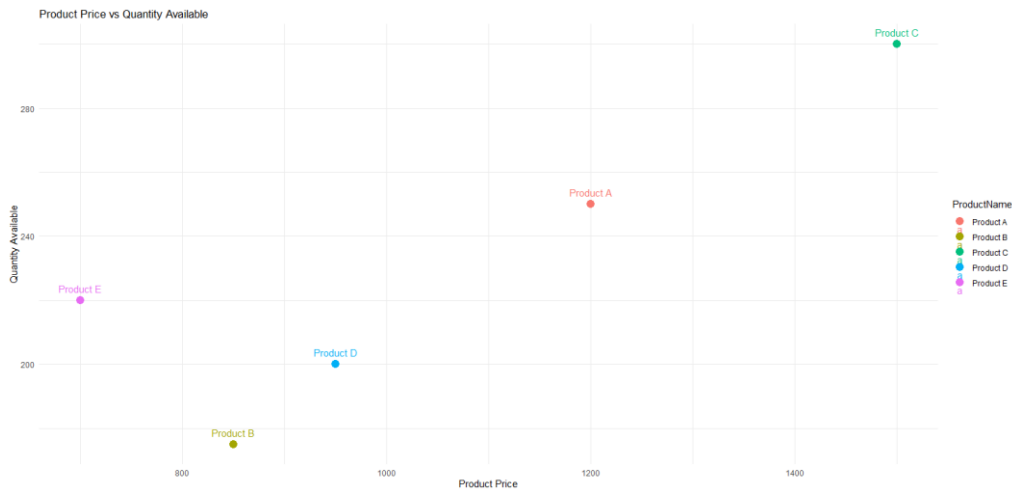
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
ggplot(inventory,  
  aes(x=Price,  
    y=Quantity,  
    color=ProductName))+  
geom_point(size=4)+  
geom_text(aes(label=ProductName),  
  vjust=-1)+  
labs(  
  title="Product Price vs Quantity Available",  
  x="Product Price",  
  y="Quantity Available"  
) +  
theme_minimal()
```

```
print("Insight:")
```

```
print("The scatter plot shows that product price and quantity available do not have a strong correlation.")
```

```
print("Some high-priced products have high stock levels, while some lower-priced products have fewer units available.")
```



4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

```
par(mfrow=c(1,2))
```

```
# Bar Chart
```

```
barplot(  
  inventory$Quantity,  
  names.arg=inventory$ProductName,  
  col=rainbow(5),  
  main="Product Inventory",  
  xlab="Product",  
  ylab="Quantity Available"  
)
```

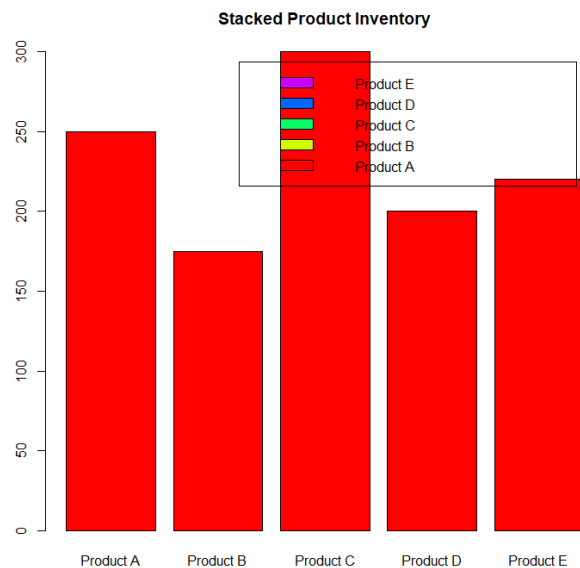
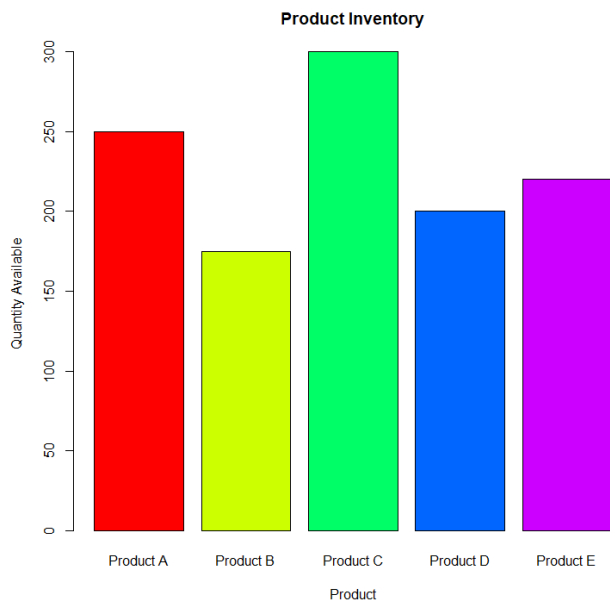
```
# Stacked Bar Chart
```

```
quantity_matrix <- matrix(  
  inventory$Quantity,  
  nrow=1,  
  byrow=TRUE  
)
```

```

colnames(quantity_matrix) <- inventory$ProductName
rownames(quantity_matrix) <- "Inventory"
barplot(
  quantity_matrix,
  beside=FALSE,
  col=rainbow(5),
  legend.text=inventory$ProductName,
  main="Stacked Product Inventory"
)
par(mfrow=c(1,1))

```



SET 5:

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
website <- data.frame(
```

```
  Date = as.Date(c("2023-01-01",
```

```
                  "2023-01-02",
```

```
                  "2023-01-03",
```

```
                  "2023-01-04",
```

```
                  "2023-01-05")),
```

```
  PageViews = c(1500,1600,1400,1650,1800),
```

```
  CTR = c(2.3,2.7,2.0,2.4,2.6)
```

```
)
```

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
ggplot(website,
```

```
  aes(x=Date,
```

```
      y=PageViews,
```

```
      group=1))+
```

```
geom_line(color="blue", linewidth=1.2)+
```

```
geom_point(color="red", size=3)+
```

```
geom_text(aes(label=PageViews),
```

```
          vjust=-1)+
```

```
labs(
```

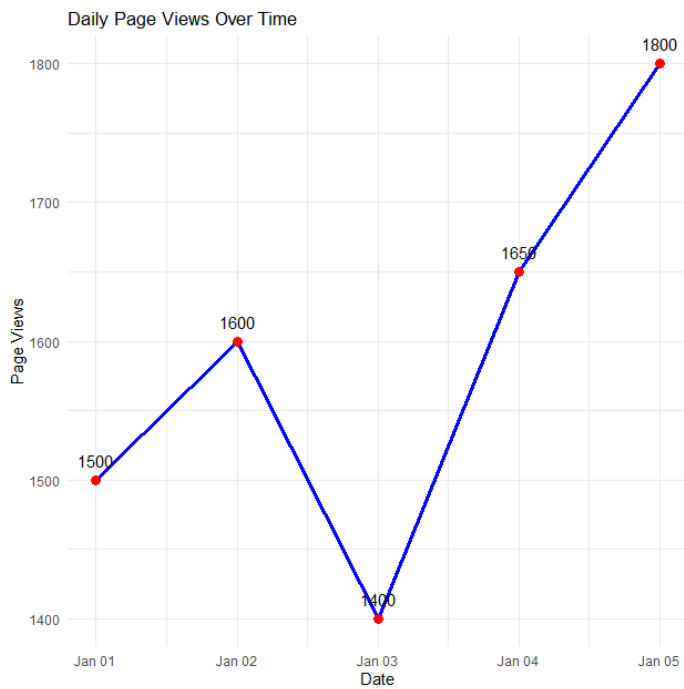
```
  title="Daily Page Views Over Time",
```

```
  x="Date",
```

```
  y="Page Views"
```

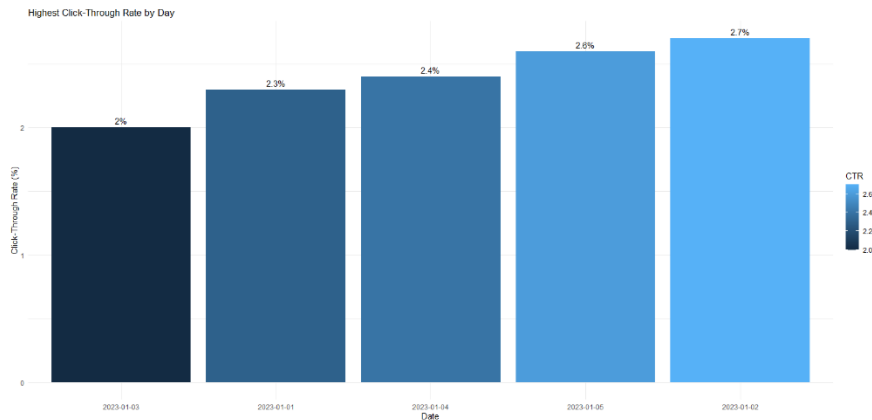
```
)+
```

```
theme_minimal()
```



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
ggplot(website,
  aes(x=reorder(as.character(Date), CTR),
    y=CTR,
    fill=CTR))+
  geom_bar(stat="identity")+
  geom_text(aes(label=paste0(CTR,"%")),
    vjust=-0.5)+
  labs(
    title="Highest Click-Through Rate by Day",
    x="Date",
    y="Click-Through Rate (%)"
  )+
  theme_minimal()
```



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

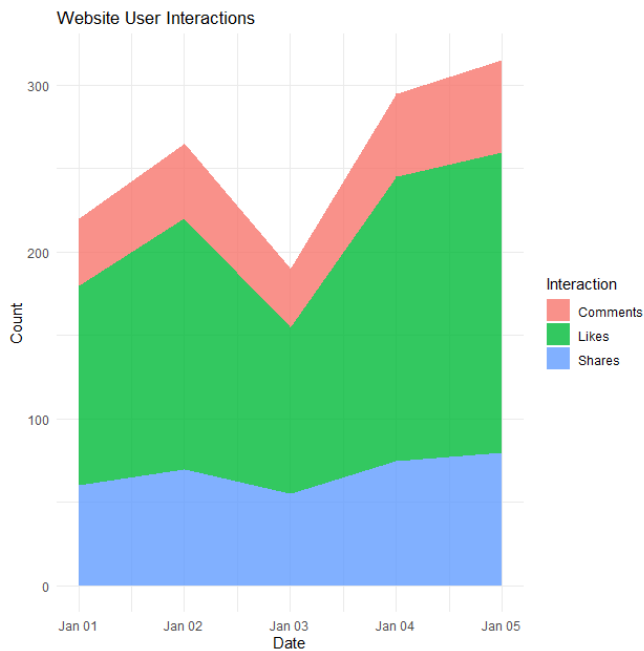
```
interaction <- data.frame(
  Date = website$Date,
  Likes = c(120,150,100,170,180),
  Shares = c(60,70,55,75,80),
  Comments = c(40,45,35,50,55)
)

interaction_long <- data.frame(
  Date = rep(interaction$Date,3),
  Type = rep(c("Likes","Shares","Comments"),
    each=nrow(interaction)),
  Count = c(interaction$Likes,
    interaction$Shares,
    interaction$Comments)
)

ggplot(interaction_long,
  aes(x=Date,
    y=Count,
    fill=Type))+

geom_area(alpha=0.8)+
```

```
labs(
  title="Website User Interactions",
  x="Date",
  y="Count",
  fill="Interaction"
)+
theme_minimal()
```



4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

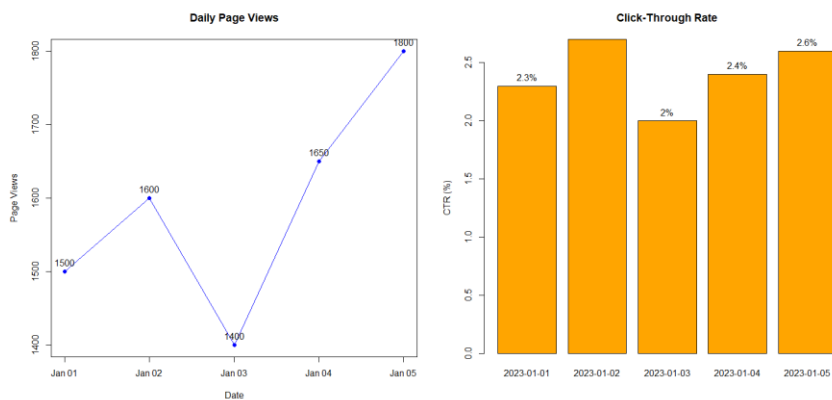
```
par(mfrow=c(1,2))
plot(
  website$Date,
  website$PageViews,
  type="o",
  col="blue",
  pch=16,
  xlab="Date",
  ylab="Page Views",
  main="Daily Page Views"
)
```



```

website$Date,
website$PageViews,
labels=website$PageViews,
pos=3
)
# Bar Chart
bar <- barplot(
  website$CTR,
  names.arg=website$Date,
  col="orange",
  main="Click-Through Rate",
  ylab="CTR (%)"
)
text(
  bar,
  website$CTR,
  labels=paste0(website$CTR,"%"),
  pos=3
)
par(mfrow=c(1,1))

```



SET 6:

```
install.packages("ggplot2")
```

```
install.packages("tidyr")
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(
```

```
  ProductID = c(1,2,3),
```

```
  ProductName = c("Product A","Product B","Product C"),
```

```
  January = c(2000,1500,1200),
```

```
  February = c(2200,1800,1400),
```

```
  March = c(2400,1600,1100)
```

```
)
```

```
sales_long <- pivot_longer(
```

```
  sales,
```

```
  cols = January:March,
```

```
  names_to = "Month",
```

```
  values_to = "Sales"
```

```
)
```

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
ggplot(sales_long,
```

```
  aes(x=ProductName,
```

```
  y=Sales,
```

```
  fill=Month))+
```

```
geom_bar(
```

```
  stat="identity",
```

```
  position="dodge"
```

```
)+
```

```
geom_text(
```

```
  aes(label=Sales),
```

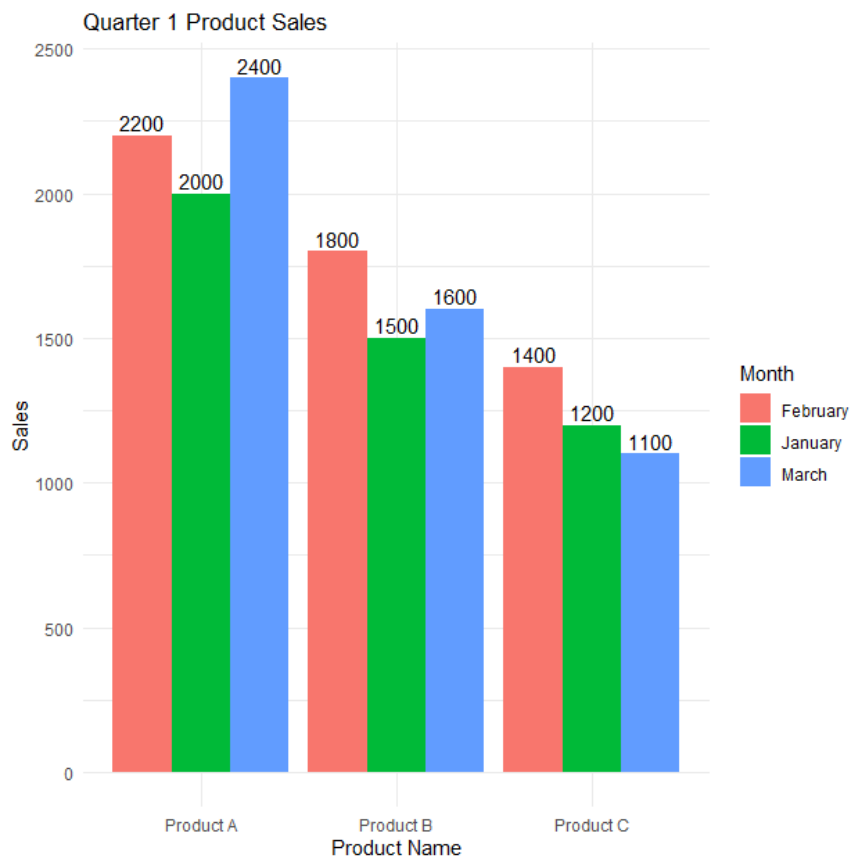
```
  position=position_dodge(width=0.9),
```

```
  vjust=-0.3
```

```

size=4
)+
labs(
  title="Quarter 1 Product Sales",
  x="Product Name",
  y="Sales",
  fill="Month"
)+
theme_minimal()

```



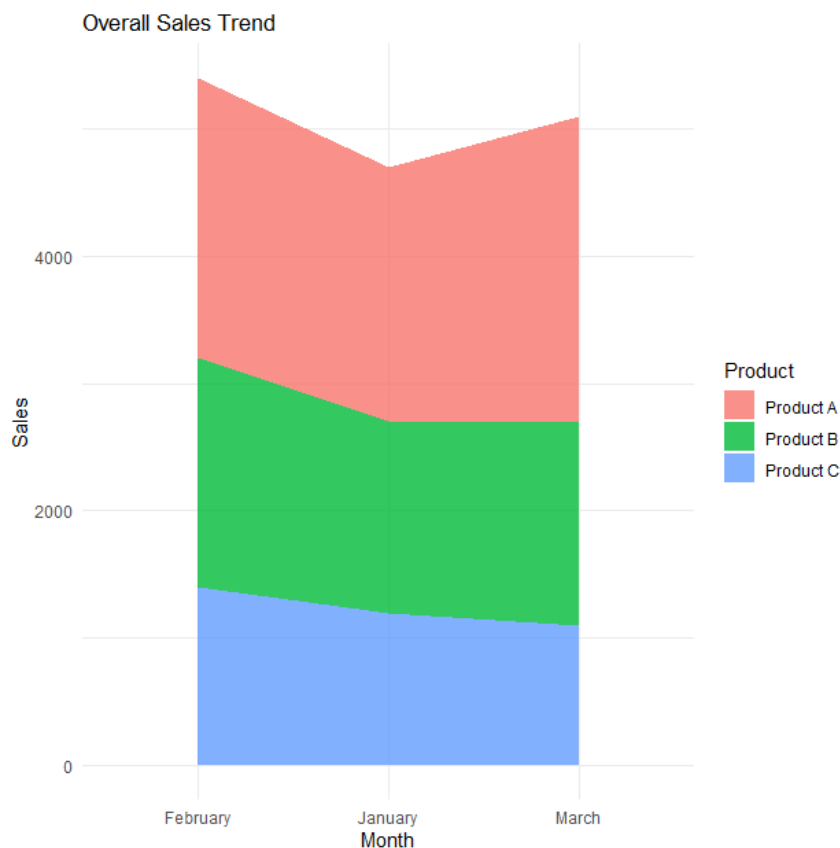
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```

ggplot(sales_long,
  aes(x=Month,
    y=Sales,
    fill=ProductName,
    group=ProductName))+

```

```
geom_area(alpha=0.8)+
labs(
  title="Overall Sales Trend",
  x="Month",
  y="Sales",
  fill="Product"
)+
theme_minimal()
```



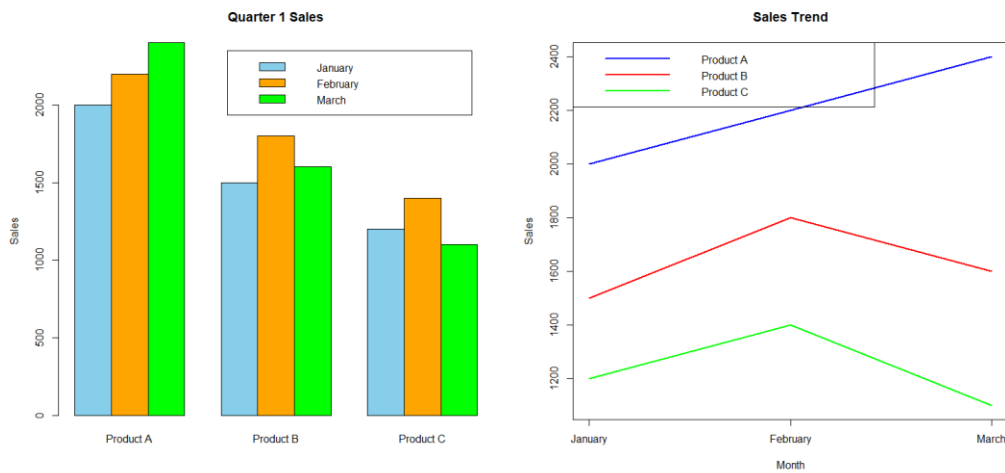
3. Build a table to show the monthly sales data for each product. Label the table elements.

```
print("Monthly Product Sales Table")
print(sales)
```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```
par(mfrow=c(1,2))
barplot(
  as.matrix(t(sales[,3:5])),
  beside=TRUE,
```

```
col=c("skyblue","orange","green"),
names.arg=sales$ProductName,
legend.text=c("January","February","March"),
main="Quarter 1 Sales",
ylab="Sales"
)
matplot(
  t(sales[,3:5]),
  type="l",
  lwd=2,
  lty=1,
  col=c("blue","red","green"),
  xaxt="n",
  xlab="Month",
  ylab="Sales",
  main="Sales Trend"
)
axis(
  1,
  at=1:3,
  labels=c("January","February","March")
)
legend(
  "topleft",
  legend=sales$ProductName,
  col=c("blue","red","green"),
  lty=1,
  lwd=2
)
par(mfrow=c(1,1))
print(sales)
```



SET 7:

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
```

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
ggplot(customer,
  aes(x=factor(CustomerID),
    y=Age,
    fill=factor(CustomerID))) +
  geom_bar(stat="identity") +
  geom_text(aes(label=Age),
    vjust=-0.5) +
  labs(
    title="Customer Age Distribution",
    x="Customer ID",
    y="Age"
  ) +
```

```
theme_minimal()+
theme(legend.position="none")
```



2. Generate a pie chart to represent the distribution of customers by gender.

```
gender <- as.data.frame(table(customer$Gender))
```

```
colnames(gender) <- c("Gender", "Count")
```

```
ggplot(gender,
```

```
  aes(x="",
```

```
    y=Count,
```

```
    fill=Gender))+
```

```
geom_bar(stat="identity", width=1)+
```

```
coord_polar("y")+
```

```
geom_text(
```

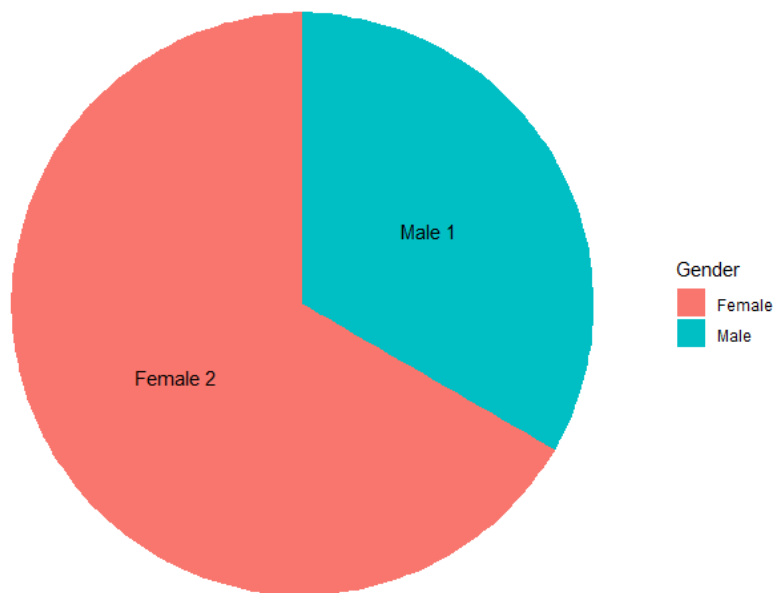
```
  aes(label=paste(Gender, Count)),
```

```
  position=position_stack(vjust=0.5)
```

```
)+
```

```
labs(
  title="Customer Gender Distribution"
)+
theme_void()
```

Customer Gender Distribution



3. Build a table to show the demographic information for each customer. Label the table elements.

```
print("Customer Demographic Information")
print(customer)
```

```
CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35  Male  60000
3          3  42 Female  75000
```

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

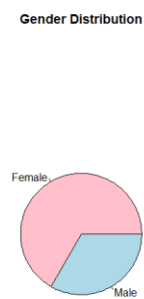
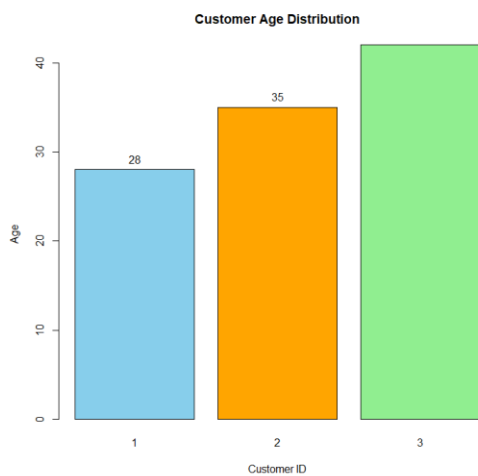
```
par(mfrow=c(1,2))
bar <- barplot(
  customer$Age,
  names.arg=customer$CustomerID,
  col=c("skyblue","orange","lightgreen"),
  main="Customer Age Distribution",
```



```

xlab="Customer ID",
ylab="Age"
)
text(
  bar,
  customer$Age,
  labels=customer$Age,
  pos=3
)
pie(
  table(customer$Gender),
  col=c("pink","lightblue"),
  main="Gender Distribution"
)
par(mfrow=c(1,1))
print(customer)

```



```

CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35  Male  60000
3          3  42 Female  75000

```